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CLOUD VS IN-HOUSE COMPUTING

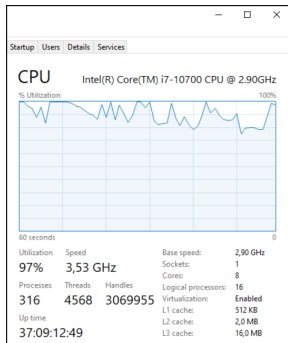
JC Huskisson



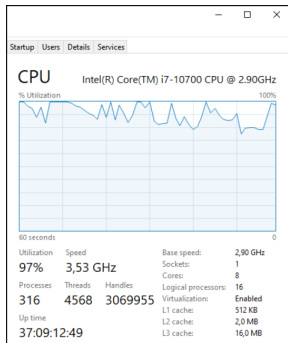
Stellenbosch Unit for Operations Research in Engineering
Department of Industrial Engineering

7 April, 2025

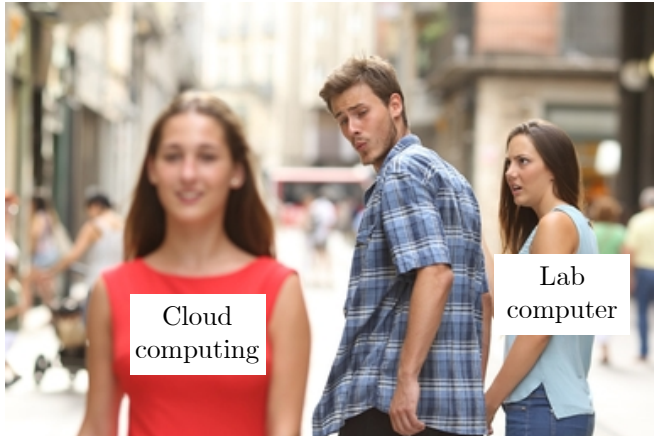
Overview



Overview



Overview



Agenda

- Solutions to computing problems

Agenda

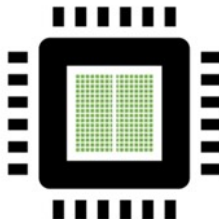
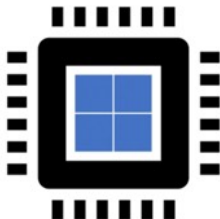
- Solutions to computing problems
- Computing time test

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- Benefits of cloud computing

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- AWS tutorial

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- Benefits of cloud computing
- AWS tutorial
- HPC overview

Two computing paradigms



CPU	GPU
Central Processing Unit	Graphics Processing Unit
4-8 Cores	100s or 1000s of Cores
Low Latency	High Throughput
Good for Serial Processing	Good for Parallel Processing
Quickly Process Tasks That Require Interactivity	Breaks Jobs Into Separate Tasks To Process Simultaneously
Traditional Programming Are Written For CPU Sequential Execution	Requires Additional Software To Convert CPU Functions to GPU Functions for Parallel Execution

Problem: CPU constrained computing

```
1  import random
2
3  def append_to_list(lst, num_items):
4      """
5      Appends num_items integers within the range [0-20000000) to the input lst
6      """
7      for n in random.sample(range(20000000), num_items):
8          print(n)
9          lst.append(n)
10
11  if __name__ == "__main__":
12      for i in range(50):
13          append_to_list([], 10000000)
14
15
```

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- Simple CPU operation of appending 10 000 000 random integers to a list

Possible solution?

- Multi CPU computing (Multiprocessing)
<https://docs.python.org/3/library/multiprocessing.html>

Problem: Computer vision

```
=====
Total params: 119,772,809
Trainable params: 119,772,809
Non-trainable params: 0
-----
Input size (MB): 9.19
Forward/backward pass size (MB): 1839.75
Params size (MB): 456.90
Estimated Total Size (MB): 2305.83
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Possible solution?

- Use *graphical processing units* (GPUs)
- Similar to multiprocessing since GPUs perform multiple simultaneous processes
- Deep learning frameworks are optimised to use GPUs without the need for manual multiprocessing.

Challenge with using many cores and GPUs?

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Intel Core i7-10700 Processor

4,8 ★★★★★ 291

Eight-core · Overclocking

R 5 899,00

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[Compare prices from 5+ shops](#)

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RTX 3050 Laptop With 64GB
RAM & 512GB SSD by Evetechn

Windows OS · Hexa Core

R 31 299,00

[Evetechn.co.za](https://www.evetechn.co.za)

R 99,00 delivery

What is cloud computing?

What is cloud computing?

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence — over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale.”

Cloud computing: Benefits

- Cost: Lower cost due to the elimination of capital and maintenance cost of hardware

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- Reliability: Cloud resources are available 24/7, regardless of loadshedding, and typically has safe cloud storage available
- Security: Cloud services has policies and technologies to keep data secure.

Computing time test

- Lab desktop:
 - Intel(R) Core i7-10700 CPU @ 2.90 GHz
- Cores: 8
Threads: 16

Computing time test

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 - Intel(R) Core i7-10700 CPU @ 2.90 GHz Cores: 8
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- Personal Laptop:
 - Intel(R) Core i5-7300 CPU @ 2.50 GHz Cores: 2
Threads: 4
 - NVIDIA GeForce GTX 1050 4GB CUDA Cores: 768

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24 GB of GDDR5 memory

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24 GB of GDDR5 memory
- Stellenbosch HPC Cluster:
 - ~~Intel Xeon Gold 5218~~
 - NVIDIA Quadro RTX 4000 8GB
 - ~~NVIDIA Quadro RTX 6000 23GB~~
 - ~~NVIDIA Tesla T4 16GB~~

Computing time test

Average iterations/s per epoch

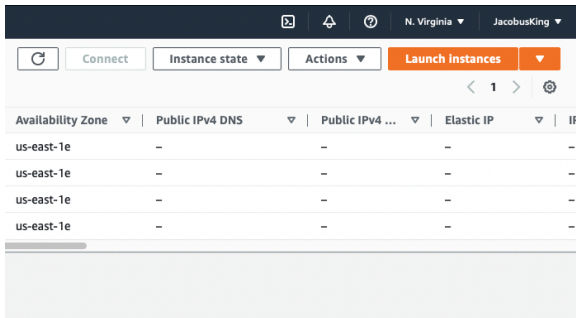
	Epoch 1	Epoch 2	Epoch 3	Epoch 4	Epoch 5	Avg
Lab computer (CPU)	0.63	0.63	0.63	0.62	0.63	0.63
Personal Laptop (CPU)	0.99	0.98	0.95	0.96	0.98	0.97
Google Colab (CPU)	1.28	1.29	1.29	1.30	1.31	1.29
Personal Laptop (GPU - GTX 1050)	7.46	7.59	7.58	7.63	7.62	7.58
Google Colab (GPU - Tesla K80)	8.25	8.26	8.26	8.25	8.25	8.25
HPC cluster (GPU - RTX 6000)	31.01	35.45	34.89	35.96	36.27	34.72

N/A

- Login to AWS

- Login to AWS
- Navigate to the EC2 service

- Login to AWS
- Navigate to the EC2 service
- Launch instance



- Name the instance
- Select an AMI

Launch an instance [info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents

Quick Start



Browse more AMIs
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type
ami-0d6cb1efb550277af (64-bit x86) / ami-0cd7323ab3e63805f (64-bit ARM)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

- Select instance type (memory, CPU, and storage)
- Create new key pair

▼ Instance type [Info](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory

On-Demand Windows pricing: 0.0162 USD per Hour

On-Demand SUSE pricing: 0.0116 USD per Hour

On-Demand RHEL pricing: 0.0716 USD per Hour

On-Demand Linux pricing: 0.0116 USD per Hour

▼

[Compare instance types](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

▼

↻

[Create new key pair](#)

- Create key pair
- Download key pair

Create key pair

×

Key pairs allow you to connect to your instance securely.

Enter the name of the key pair below. When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Key pair name

GS

The name can include upto 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair (Not supported for Windows instances)

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel Create key pair

● Launch instance

▼ Summary

Number of instances [Info](#)

1

[Software image \(AMI\)](#)
Amazon Linux 2 Kernel 5.10 AMI...[read more](#)
ami-0dfcb1ef8550277af

[Virtual server type \(instance type\)](#)
t2.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

④ **Free tier:** In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

×

Cancel [Launch Instance](#)

- Go to instance page
- Select instance
- Connect

Instances (1/5) Info									Connect
Find instance by attribute or tag (case-sensitive)									
	Name ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status	Availability Zone ▾		
☐	Generic_Skills	i-0cb6ac715e4d72d91	 Terminated  	t2.micro	-	No alarms 	us-east-1e		
☐	G5	i-0fa54c96742233198	 Terminated  	t2.micro	-	No alarms 	us-east-1e		
☐	Test	i-02e8bd4d4fa31e2a0	 Terminated  	t2.micro	-	No alarms 	us-east-1e		
☒	G5	i-08cfcfb40e3fca81	 Running  	t2.micro	 Initializing	No alarms 	us-east-1e		
☐	G5	i-0bb2c0acef3d5c407	 Terminated  	t2.micro	-	No alarms 	us-east-1e		

● SSH Client

Connect to instance [Info](#)
Connect to your instance `i-08cfcfb40e3fcaf81` (GS) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID
`i-08cfcfb40e3fcaf81` (GS)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is `GS.pem`
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 GS.pem`
4. Connect to your Instance using its Public DNS:
`ec2-3-80-18-72.compute-1.amazonaws.com`

Example:
`ssh -i "GS.pem" ec2-user@ec2-3-80-18-72.compute-1.amazonaws.com`

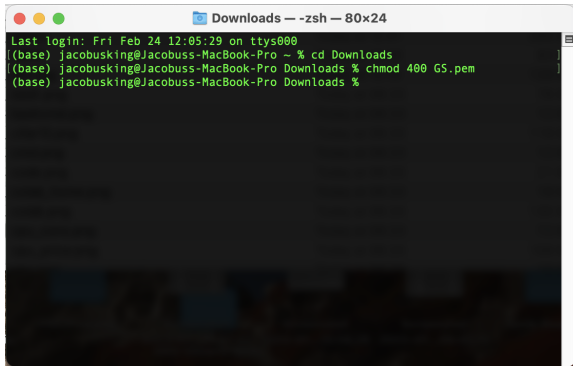
Note: In most cases, the guessed user name is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

AWS Tutorial (Windows)

- The following slides are for mac users
- The commands are similar for Windows users, however, Putty should be installed
- Download Putty
- Windows instructions

AWS Tutorial (Mac)

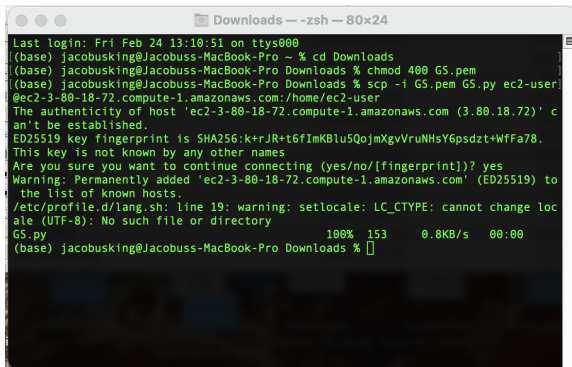
- Open terminal
- Navigate to Downloads (or where the key pair is stored):
cd Downloads
- Run in terminal: `chmod 400 [key pair name].pem`



```
Downloads — -zsh — 80x24
Last login: Fri Feb 24 12:05:29 on ttys000
(base) jacobusking@Jacobuss-MacBook-Pro ~ % cd Downloads
(base) jacobusking@Jacobuss-MacBook-Pro Downloads % chmod 400 GS.pem
(base) jacobusking@Jacobuss-MacBook-Pro Downloads %
```

AWS Tutorial (Mac)

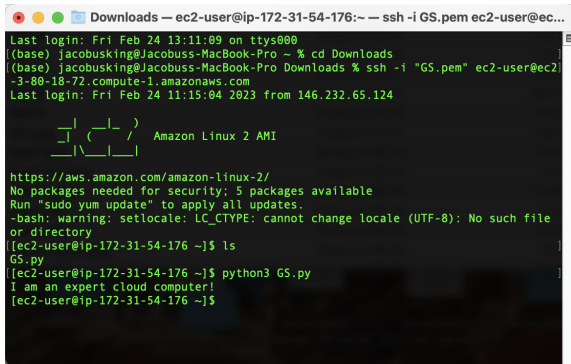
- With python file in the same folder, run in terminal: `scp -i [key pair name].pem [python file name].py ec2-user@[public DNS of instance]:/home/ec2-user`
- Input “yes”

A screenshot of a macOS terminal window titled "Downloads — -zsh — 80x24". The terminal shows a series of commands and their outputs. The user navigates to the Downloads directory, changes permissions on a file named GS.pem, and then runs an SCP command to copy GS.pem to an Amazon EC2 instance. The terminal displays a warning about the host's authenticity, showing the host's fingerprint (SHA256:k+rJR+t6fImKBlu5QojmXgvVruNHsY6psdzt+WfFa78). The user responds with 'yes' to continue the connection. The terminal also shows a warning from the system about the LC_CTYPE environment variable and a progress bar for the file transfer.

```
Last login: Fri Feb 24 13:10:51 on ttys000
(base) jacobusking@Jacobuss-MacBook-Pro ~ % cd Downloads
(base) jacobusking@Jacobuss-MacBook-Pro Downloads % chmod 400 GS.pem
(base) jacobusking@Jacobuss-MacBook-Pro Downloads % scp -i GS.pem GS.py ec2-user@ec2-3-80-18-72.compute-1.amazonaws.com:/home/ec2-user
The authenticity of host 'ec2-3-80-18-72.compute-1.amazonaws.com (3.80.18.72)' can't be established.
ED25519 key fingerprint is SHA256:k+rJR+t6fImKBlu5QojmXgvVruNHsY6psdzt+WfFa78.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-80-18-72.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
/etc/profile.d/lang.sh: line 19: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file or directory
GS.py                               100% 153      0.8KB/s   00:00
(base) jacobusking@Jacobuss-MacBook-Pro Downloads %
```

AWS Tutorial (Mac)

- To connect to instance, run in terminal: `ssh -i "[key pair name].pem" ec2-user@[public DNS of instance]`
- You are now connected to the instance
- To view your file on the instance, run: `ls`
- Run your python file: `python3 [python file name].py`



```
Downloads — ec2-user@ip-172-31-54-176:~ — ssh -i GS.pem ec2-user@ec...
Last login: Fri Feb 24 13:11:09 on ttys000
(base) jacobusking@Jacobuss-MacBook-Pro ~ % cd Downloads
(base) jacobusking@Jacobuss-MacBook-Pro Downloads % ssh -i "GS.pem" ec2-user@ec2
-3-80-18-72.compute-1.amazonaws.com
Last login: Fri Feb 24 11:15:04 2023 from 146.232.65.124

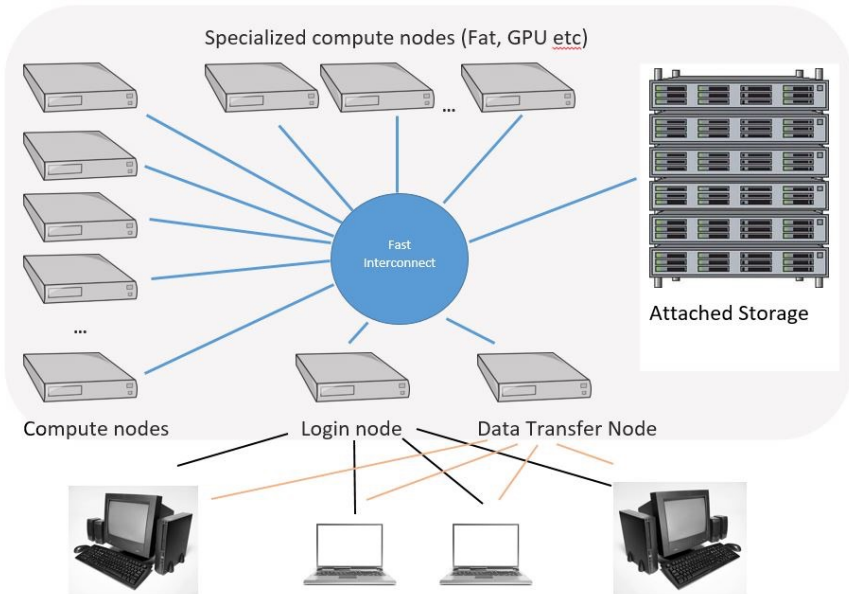
  _| _|_ )
  _| ( _| /   Amazon Linux 2 AMI
  _|\_|_|_|

https://aws.amazon.com/amazon-linux-2/
No packages needed for security; 5 packages available
Run "sudo yum update" to apply all updates.
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
[ec2-user@ip-172-31-54-176 ~]$ ls
GS.py
[ec2-user@ip-172-31-54-176 ~]$ python3 GS.py
I am an expert cloud computer!
[ec2-user@ip-172-31-54-176 ~]$
```

Steps to use HPC cluster



Steps to use HPC cluster



Steps to use HPC cluster

- Connect to cluster
- Create environment file
- Set up bash script
- Send data to cluster
- Submit job
- Monitor job.

Notes before we begin

Notes before we begin

- Must be connected to Stellenbosch LAN network, Eduroam does not work
- Be considerate and read through the docs
 - HPC website
 - HPC graphics card cluster github
- Each person gets 1 000 free hours to test the HPC
- Remember to cite HPC
- **DO NOT USE IT FOR PERSONAL PROJECTS.**

Connect to cluster: Mac

Connect to cluster: Mac

- 1 Open terminal
- 2 type “<student number>@hpc1.sun.ac.za”
- 3 type “yes” if prompted
- 4 type password when prompted (password is not visible).

```
(base) → ssh 20706677@hpc1.sun.ac.za
The authenticity of host 'hpc1.sun.ac.za (146.232.129.166)' can't be established.
ED25519 key fingerprint is SHA256:B1FVYRHJA+bcu1XnMuPYN/pNYcli2chlyM/JR1146jc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'hpc1.sun.ac.za' (ED25519) to the list of known hosts.
(20706677@hpc1.sun.ac.za) Password:
Last login: Wed Mar  2 13:04:54 2022 from 146.232.28.92

*****
HPC MAINTENANCE 10 - 14 JAN 2022 concluded OK.  Jobs should run!

See http://www.sun.ac.za/hpc for details on how to use the HPC.

Access to this system is governed by the Electronic Communication Policy (http://stbweb01.stb.sun.ac.za/ecp/ecp\_december\_eng.pdf)

Contact IT Helpdesk <help@sun.ac.za> for assistance.
Research Help Desk Call logging:
https://servicedesk.sun.ac.za/jira/plugins/servlet/theme/portal/22/group/92

*****

Remember to acknowledge the HPC in your publications. For detail on
how to do this, see http://www.sun.ac.za/hpc#Citations

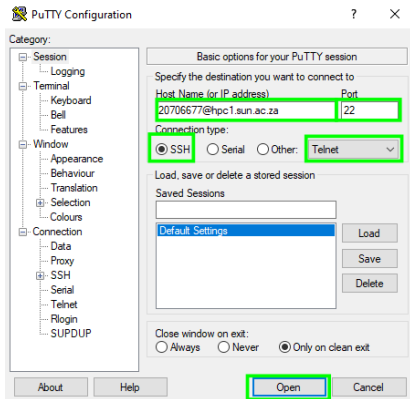
*****

-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file or directory
[20706677@hpc1 ~]$
```

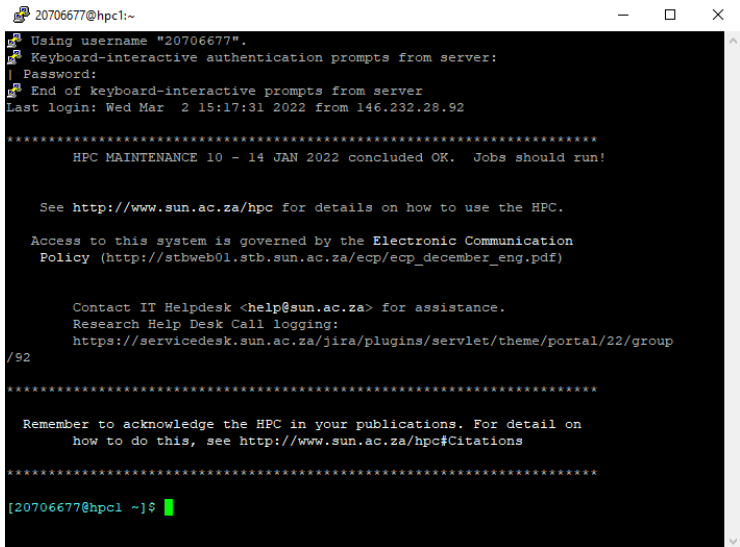
Connect to cluster: Windows

Connect to cluster: Windows

- 1 Download and open open Putty
- 2 select SSH, Telnet, set port = 22, and type “<student number>@hpc1.sun.ac.za” in Host Name
- 3 click on open and type ”yes” if prompted
- 4 type password when prompted (password is not visible).



Connect to cluster: Windows



A terminal window titled '20706677@hpc1:~' with standard window controls. The terminal displays the following text:

```
Using username "20706677".
Keyboard-interactive authentication prompts from server:
| Password:
End of keyboard-interactive prompts from server
Last login: Wed Mar  2 15:17:31 2022 from 146.232.28.92

*****
      HPC MAINTENANCE 10 - 14 JAN 2022 concluded OK.  Jobs should run!

      See http://www.sun.ac.za/hpc for details on how to use the HPC.

      Access to this system is governed by the Electronic Communication
      Policy (http://stbweb01.stb.sun.ac.za/ecp/ecp\_december\_eng.pdf)

      Contact IT Helpdesk <help@sun.ac.za> for assistance.
      Research Help Desk Call logging:
      https://servicedesk.sun.ac.za/jira/plugins/servlet/theme/portal/22/group
/92

*****

      Remember to acknowledge the HPC in your publications. For detail on
      how to do this, see http://www.sun.ac.za/hpc#Citations

*****

[20706677@hpc1 ~]$
```

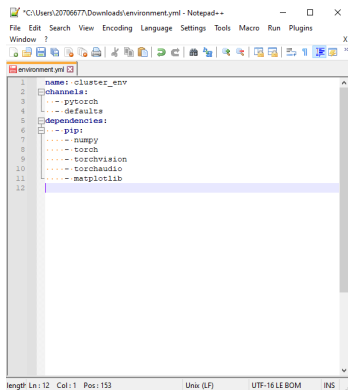
Connect to cluster: Windows

- ~~Connect to cluster~~
- Create environment file
- Set up bash script
- Send data to cluster
- Submit job
- Monitor job.

Create environment file

Create an environment file with the required packages.

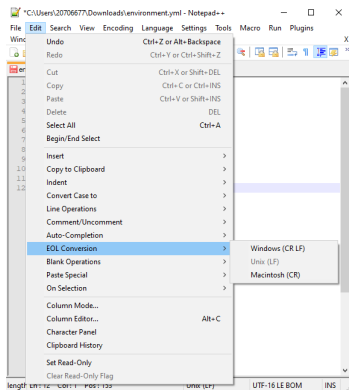
Windows: Do this in notepad++ and ensure that the *end of line* (EOL) conversion is set to UNIX (LF)



The screenshot shows a Notepad++ window titled "environment.yml - Notepad++". The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, and Plugins. The toolbar contains various icons for file operations and editing. The text area displays the following content:

```
1 name: cluster_env
2 channels:
3   - pytorch
4   - defaults
5 dependencies:
6   - pip:
7     - numpy
8     - torch
9     - torchvision
10    - torchaudio
11    - matplotlib
12
```

The status bar at the bottom indicates "length Ln: 12 Col: 1 Pos: 153", "Unix (LF)", "UTF-16 LE BOM", and "INS".



The screenshot shows the same Notepad++ window with the "Edit" menu open. The "EOL Conversion" option is highlighted, and a submenu is displayed with the following options:

- Windows (CR LF)
- Unix (LF)
- Macintosh (CR)

The status bar at the bottom indicates "length Ln: 12 Col: 1 Pos: 153", "Unix (LF)", "UTF-16 LE BOM", and "INS".



TIP: Export all manually installed Conda packages with ``conda env export --from-history``

Create environment file

- ~~Connect to cluster~~
- ~~Create environment file~~
- Set up bash script
- Send data to cluster
- Submit job
- Monitor job.

Set up bash script

- Create a new file and copy the bash script information from the HPC-GPU website
- Adapt bash script details to match your scenario:
 - Change jobname
 - Select resources
 - Change walltime
 - Provide email for termination notification
 - Ensure path to python file is correct
- Save the bash file as <filename>.sh in your code folder.

Set up bash script

```
environment.yml  new 1 x
1  #!/bin/bash
2  #PBS -N JobName
3  #PBS -l select=1:ncpus=4:mem=16GB:ngpus=1:Qlist=ee
4  #PBS -l walltime=6:00:00
5  #PBS -m ae
6  #PBS -e output.err
7  #PBS -o output.out
8  #PBS -M <username>@sun.ac.za
9
10 # make sure I'm the only one that can read my output
11 umask 0077
12 # create a temporary directory with the job ID as name in /scratch-small-local
13 SPACED="${PBS_JOBID//./-}"
14 TMP=/scratch-small-local/${SPACED} # E.g. 249926.hpcl.hpc
15 mkdir -p ${TMP}
16 echo "Temporary work dir: ${TMP}"
17
18 cd ${TMP}
19
20 wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O miniconda.sh
21 bash miniconda.sh -b -p miniconda
22
23 # Ensure miniconda is activated
24 echo "Activating conda"
25 source ../miniconda/bin/activate
26
27 # copy the input files to ${TMP}
28 echo "Copying from ${PBS_O_WORKDIR} to ${TMP}/"
29 /usr/bin/rsync -vax "${PBS_O_WORKDIR}/" "${TMP}/"
30
31 # write my output to my new temporary work directory
32 conda env create -f environment.yml
33 export LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${TMP}/miniconda/lib/
34 conda activate <env name>
35 # You may need to add additional lines here if your script
36 # requires custom git / pip dependancies not included in your conda env.
37 python train.py <args>
38
39 # job done, copy everything back
40 echo "Copying from ${TMP}/ to ${PBS_O_WORKDIR}/"
41 /usr/bin/rsync -vax ${TMP}/ "${PBS_O_WORKDIR}/"
42
43 # if the copy back succeeded, delete my temporary files
```

Set up bash script

- ~~Connect to cluster~~
- ~~Create environment file~~
- ~~Set up bash script~~
- Send data to cluster
- Submit job
- Monitor job.

Prepare to send data (Windows)

In order to open a Linux terminal in at the desired location, it is first needed to acquire a Linux terminal. (Full tutorial <https://itsfoss.com/install-bash-on-windows/>)

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In short:

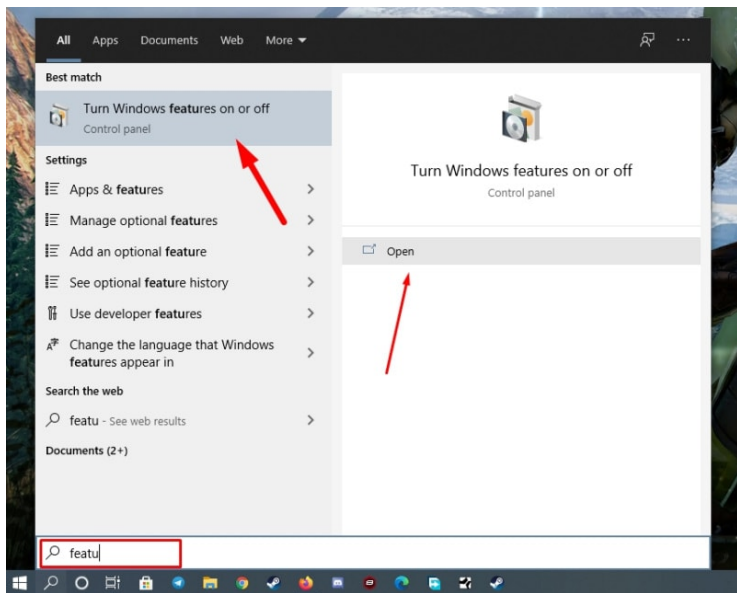
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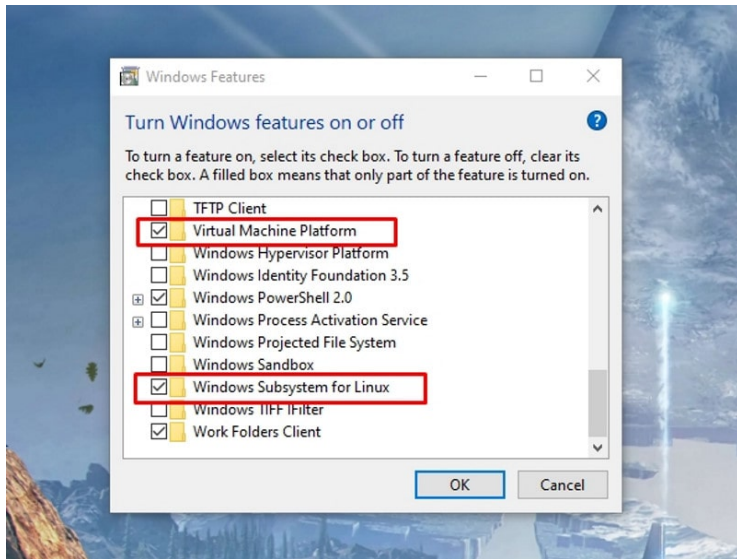
In short:

- Turn on virtual machine and windows subsystem features

Prepare to send data (Windows)



Prepare to send data (Windows)



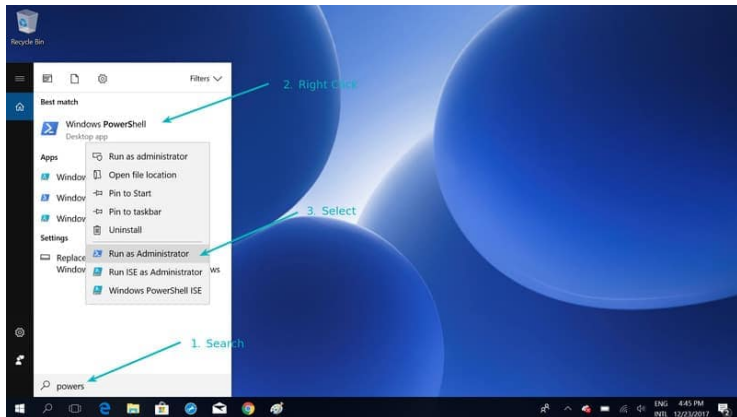
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In short:

- Turn on virtual machine and windows subsystem features
- To enable Bash, open windows power shell as administrator, and run command shown

Prepare to send data (Windows)



```
Enable-WindowsOptionalFeature -Online -FeatureName  
VirtualMachinePlatform -norestart  
dism.exe /online /enable-feature /featurename:Microsoft-Windows-  
Subsystem-Linux /all
```

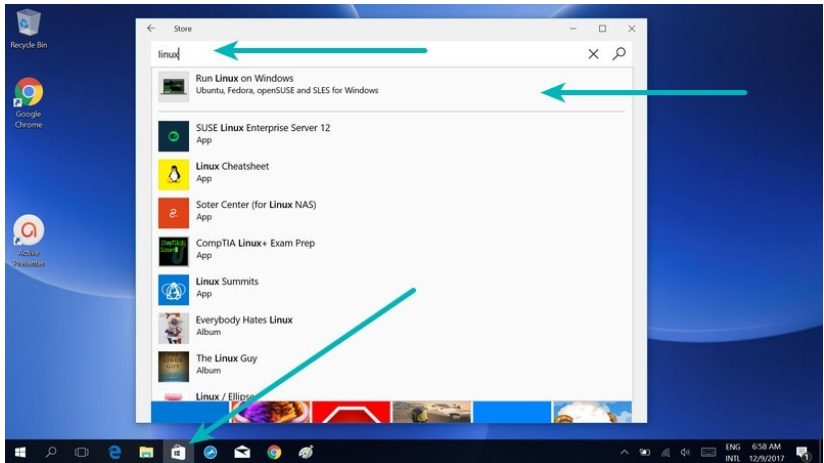
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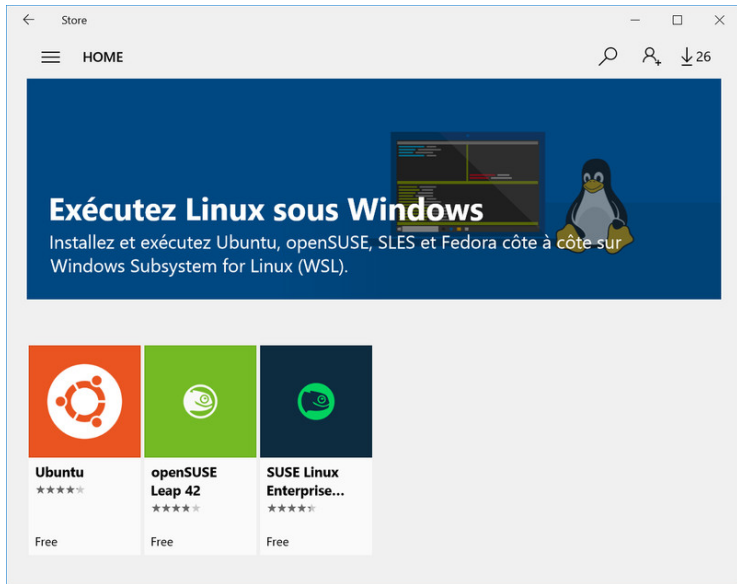
In short:

- Turn on virtual machine and windows subsystem features
- To enable Bash, open windows power shell as administrator, and run command shown
- Install Ubuntu

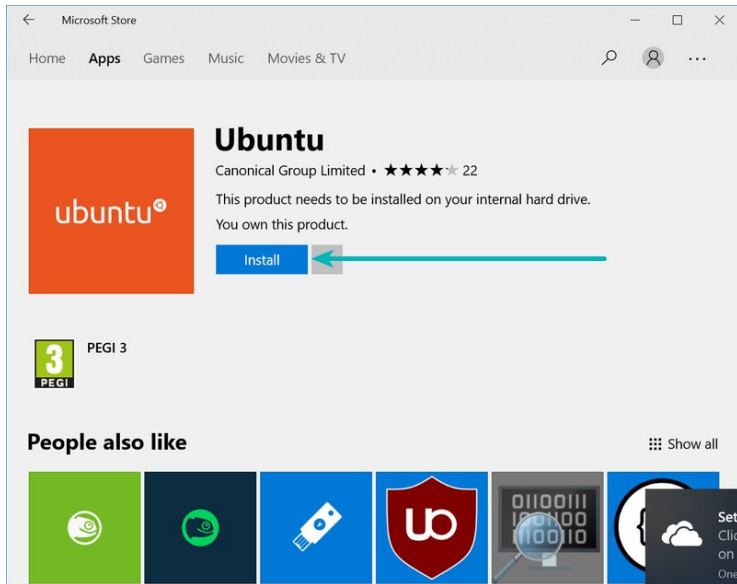
Prepare to send data (Windows)



Prepare to send data (Windows)



Prepare to send data (Windows)



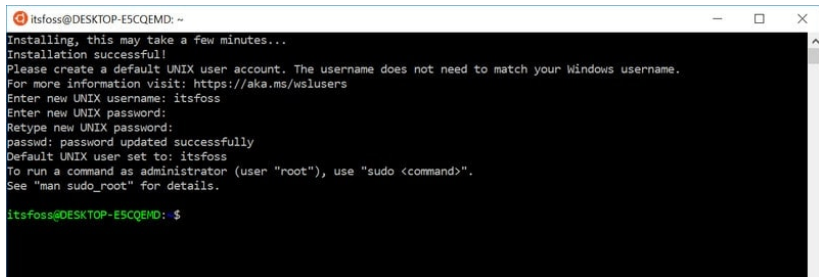
Prepare to send data (Windows)

In order to open a Linux terminal in at the desired location, it is first needed to acquire a Linux terminal. (Full tutorial <https://itsfoss.com/install-bash-on-windows/>)

In short:

- Turn on virtual machine and windows subsystem features
- To enable Bash, open windows power shell as administrator, and run command shown
- Install Ubuntu
- Search and run Ubuntu. Setup UNIX with username and password

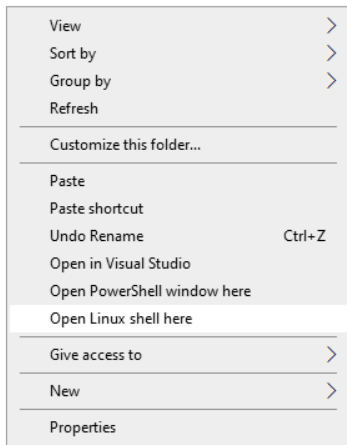
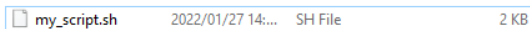
Prepare to send data (Windows)



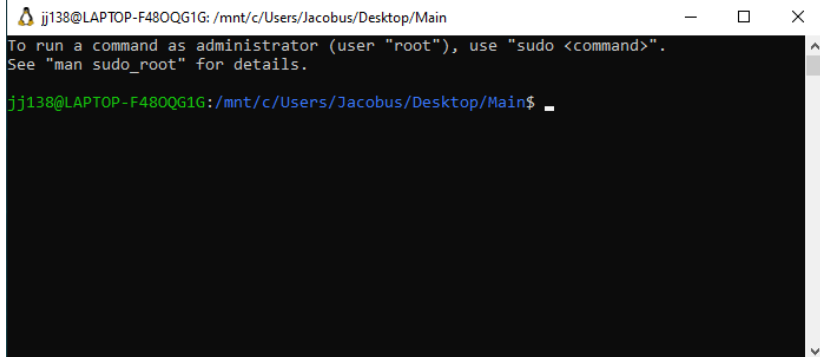
```
itsfoss@DESKTOP-E5CQEMD: ~  
Installing, this may take a few minutes...  
Installation successful!  
Please create a default UNIX user account. The username does not need to match your Windows username.  
For more information visit: https://aka.ms/wslusers  
Enter new UNIX username: itsfoss  
Enter new UNIX password:  
Retype new UNIX password:  
passwd: password updated successfully  
Default UNIX user set to: itsfoss  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
itsfoss@DESKTOP-E5CQEMD: $
```

Send data to cluster

- Open linux shell in the folder that contains the file/folder you wish to send (windows: shift + right click)



Send data to cluster



```
jj138@LAPTOP-F48OQG1G: /mnt/c/Users/Jacobus/Desktop/Main
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
jj138@LAPTOP-F48OQG1G:/mnt/c/Users/Jacobus/Desktop/Main$
```

UPLOAD

```
rsync -vax --progress <filename> <std_num>@hpc1.sun.ac.za:~/filename
rsync -vax --progress my_script.sh 20706677@hpc1.sun.ac.za:~/my_script.sh
```

DOWNLOAD

```
rsync -vax --progress <std_num>@hpc1.sun.ac.za:~/filename <filename>
rsync -vax --progress 20706677@hpc1.sun.ac.za:~/my_script.sh my_script.sh
```

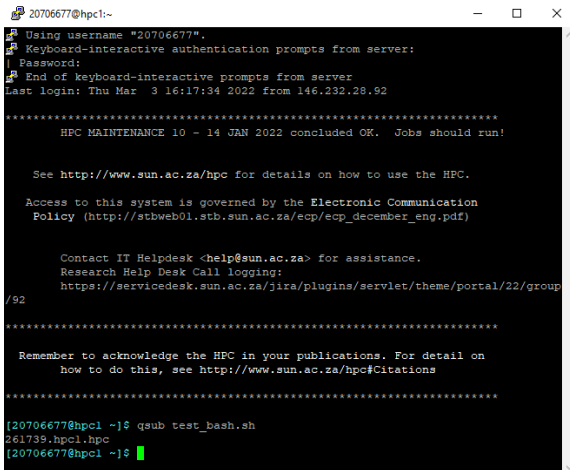
Send data to cluster

- ~~Connect to cluster~~
- ~~Create environment file~~
- ~~Set up bash script~~
- ~~Send data to cluster~~
- Submit job
- Monitor job.

Submit job

In order to submit job type in the cmd while being in the directory that contains the bash script:

- `qsub <bash_script_name>.sh`

A terminal window titled '20706677@hpc1:~' showing the login process for a user named '20706677'. The terminal displays a series of prompts and messages, including 'Using username "20706677".', 'Keyboard-interactive authentication prompts from server:', 'Password:', 'End of keyboard-interactive prompts from server', and 'Last login: Thu Mar 3 16:17:34 2022 from 146.232.28.92'. Following the login, the terminal shows a message about HPC maintenance and a link to the HPC website. The user then enters the command 'qsub test_bash.sh' and the terminal shows the output '261739.hpc1.hpc' and a green cursor.

```
20706677@hpc1:~  
Using username "20706677".  
Keyboard-interactive authentication prompts from server:  
Password:  
End of keyboard-interactive prompts from server  
Last login: Thu Mar 3 16:17:34 2022 from 146.232.28.92  
  
*****  
HPC MAINTENANCE 10 - 14 JAN 2022 concluded OK. Jobs should run!  
  
See http://www.sun.ac.za/hpc for details on how to use the HPC.  
  
Access to this system is governed by the Electronic Communication  
Policy (http://stbweb01.stb.sun.ac.za/ecp/ecp\_december\_eng.pdf)  
  
Contact IT Helpdesk <help@sun.ac.za> for assistance.  
Research Help Desk Call logging:  
https://servicedesk.sun.ac.za/jira/plugins/servlet/theme/portal/22/group/92  
  
*****  
  
Remember to acknowledge the HPC in your publications. For detail on  
how to do this, see http://www.sun.ac.za/hpc#Citations  
  
*****  
  
[20706677@hpc1 ~]$ qsub test_bash.sh  
261739.hpc1.hpc  
[20706677@hpc1 ~]$
```

- ~~Connect to cluster~~
- ~~Create environment file~~
- ~~Set up bash script~~
- ~~Send data to cluster~~
- ~~Submit job~~
- Monitor job.

- Get job number
 - Keep job number from when job was created
 - Find active jobs on node by running: `pestat`

```
comp055.hpc    free    64    0   371G   0%  
comp056.hpc    free    64   32   371G  54%  261732  
comp059.hpc    offl*    0    0    70G   0%
```

- For instant feedback of current progress:
 - `qpeek <job_num>`
- For continuous feedback of current progress:
 - `qpeek -f <job_num>`
- For snapshot feedback of current progress:
 - `qpeek <job_num> -t`

Other useful commands

- Delete a job:
qdel <job_id>

Other useful commands

- Delete a job:
`qdel <job_id>`
- See utilisation of all nodes:
`pestat`

Other useful commands

```
20706677@hpc1:~  
[20706677@hpc1 ~]$ pestat  
Queues:  short day week month long  
Node      state  cpu      memory  jobsda/users  
----  
comp002.hpc  free    64 35 126G 23% 261695  
comp022.hpc  free    64 0 126G 0%  
comp023.hpc  free    48 32 90G 71% 261685 261620  
comp024.hpc  offl*   48 0 94G 0%  
comp025.hpc  free    48 35 90G 33% 261696  
comp029.hpc  free    48 24 94G 92% 261686 261472  
comp030.hpc  free    48 0 94G 0%  
comp031.hpc  free    48 0 94G 0%  
comp032.hpc  free    48 32 94G 68% 261688 261621  
comp033.hpc  free    48 32 94G 68% 261736 261622  
comp034.hpc  free    48 32 94G 68% 261623 261738  
comp036.hpc  free    48 0 94G 0%  
comp037.hpc  free    48 0 94G 0%  
comp040.hpc  free    64 60 126G 95% 261604  
comp041.hpc  free    64 0 126G 0%  
comp042.hpc  free    64 35 363G 41% 261606  
comp043.hpc  free    64 60 363G 41% 261608  
comp044.hpc  free    64 60 363G 52% 261609  
comp045.hpc  free    64 0 363G 0%  
comp046.hpc  free    64 0 363G 0%  
comp047.hpc  free    64 0 363G 0%  
comp048.hpc  free    64 0 363G 0%  
comp049.hpc  free    64 0 363G 0%  
comp050.hpc  busy    64 64 363G 35% 261733 261734 261735 261684  
comp051.hpc  free    64 0 363G 0%  
comp052.hpc  free    48 0 1278G 0%  
comp055.hpc  free    64 0 371G 0%  
comp056.hpc  free    64 32 371G 54% 261732  
comp059.hpc  offl*   0 0 70G 0%  
comp060.hpc  free    24 0 70G 0%  
comp062.hpc  offl*   0 0 126G 0%  
comp064.hpc  free    96 90 997G 90% 261574  
comp067.hpc  free    24 0 35G 0%  
comp069.hpc  free    24 0 35G 0%  
comp070.hpc  free    24 0 35G 0%  
comp071.hpc  free    24 0 35G 0%  
comp072.hpc  offl*   0 0 0G 0%  
comp073.hpc  offl*   0 0 0G 0%  
comp074.hpc  offl*   0 0 0G 0%  
comp075.hpc  offl*   0 0 0G 0%  
comp076.hpc  free    24 0 35G 0%  
comp082.hpc  free    72 60 1510G 12% 261610  
comp084.hpc  free    64 0 376G 0%
```

Other useful commands

- Delete a job:
`qdel <job_id>`
- See utilisation of all nodes:
`pestat`
- Remove miniconda folder when miniconda errors occur:
`rm -r miniconda`

Other useful commands

- Delete a job:
`qdel <job_id>`
- See utilisation of all nodes:
`pestat`
- Remove miniconda folder when miniconda errors occur:
`rm -r miniconda`
- In order to interact with the cluster environment:
`mc`

Other useful commands

mc [20706677@hpc1.hpc]:~

Left	File	Command	Options	Right			
< ~				< ~			
[~]				[~]			
'n	Name	Size	Modify time	'n	Name	Size	Modify time
/..	UP--DIR	Feb 28 21:13		/..	UP--DIR	Feb 28 21:13	
/.cache	39	Jan 27 14:31		/.cache	39	Jan 27 14:31	
/.conda	38	Jan 26 13:15		/.conda	38	Jan 26 13:15	
/.config	24	Jan 24 12:04		/.config	24	Jan 24 12:04	
/.local	27	Jan 24 12:04		/.local	27	Jan 24 12:04	
/.mozilla	51	Jan 24 10:07		/.mozilla	51	Jan 24 10:07	
/.nv	34	Jan 27 16:29		/.nv	34	Jan 27 16:29	
/.ssh	122	Mar 2 08:24		/.ssh	122	Mar 2 08:24	
/Meesters	71	Feb 28 14:39		/Meesters	71	Feb 28 14:39	
/results	10	Feb 21 12:01		/results	10	Feb 21 12:01	
.bash_history	12783	Mar 2 14:50		.bash_history	12783	Mar 2 14:50	
.bash_logout	18	Jan 24 10:07		.bash_logout	18	Jan 24 10:07	
.bash_profile	193	Jan 24 10:07		.bash_profile	193	Jan 24 10:07	
.bashrc	231	Jan 24 10:07		.bashrc	231	Jan 24 10:07	
.kshrc	172	Jan 24 10:07		.kshrc	172	Jan 24 10:07	
UP--DIR				UP--DIR			
24T/73T (33%)				24T/73T (33%)			

Hint: Key frequently visited ftp sites in the hotlist: type C-\.
[20706677@hpc1 ~]\$ [~]

1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit

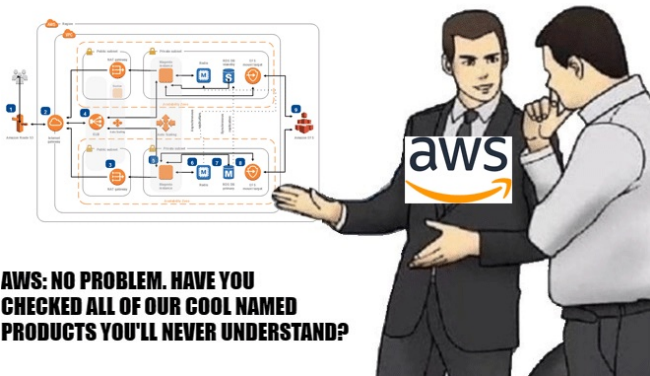
Other useful commands

- Delete a job:
`qdel <job_id>`
- See utilisation of all nodes:
`pestat`
- Remove miniconda folder when miniconda errors occur:
`rm -r miniconda`
- In order to interact with the cluster environment:
`mc`
- Transfer large datasets:
 - Compressing (Open linux terminal in directory in which data_folder is):
`tar -czvf data_folder.tar.gz data_folder`
 - Extracting (Run in the location where the data_folder.tar.gz is):
`tar -xzvf data_folder.tar.gz`

Useful links:

- [HPC cluster instructions](#)
- [HPC quickstart guide](#)

ME: I JUST NEED TO HOST 'HELLO WORLD' ON THE CLOUD.



**AWS: NO PROBLEM. HAVE YOU
CHECKED ALL OF OUR COOL NAMED
PRODUCTS YOU'LL NEVER UNDERSTAND?**